

Realtime blur simulation of varifocal spectacle lenses in virtual reality

Yannick Sauer
University of Tübingen
Tübingen, Germany
yannick.sauer@uni-tuebingen.de

Selam Wondimu Habtegiorgis
Zeiss Vision International GmbH
Aalen, Germany
selam-
wondimu.habtegiorgis@zeiss.com

Siegfried Wahl
University of Tübingen
Tübingen, Germany
Zeiss Vision International GmbH
Aalen, Germany
siegfried.wahl@uni-tuebingen.de

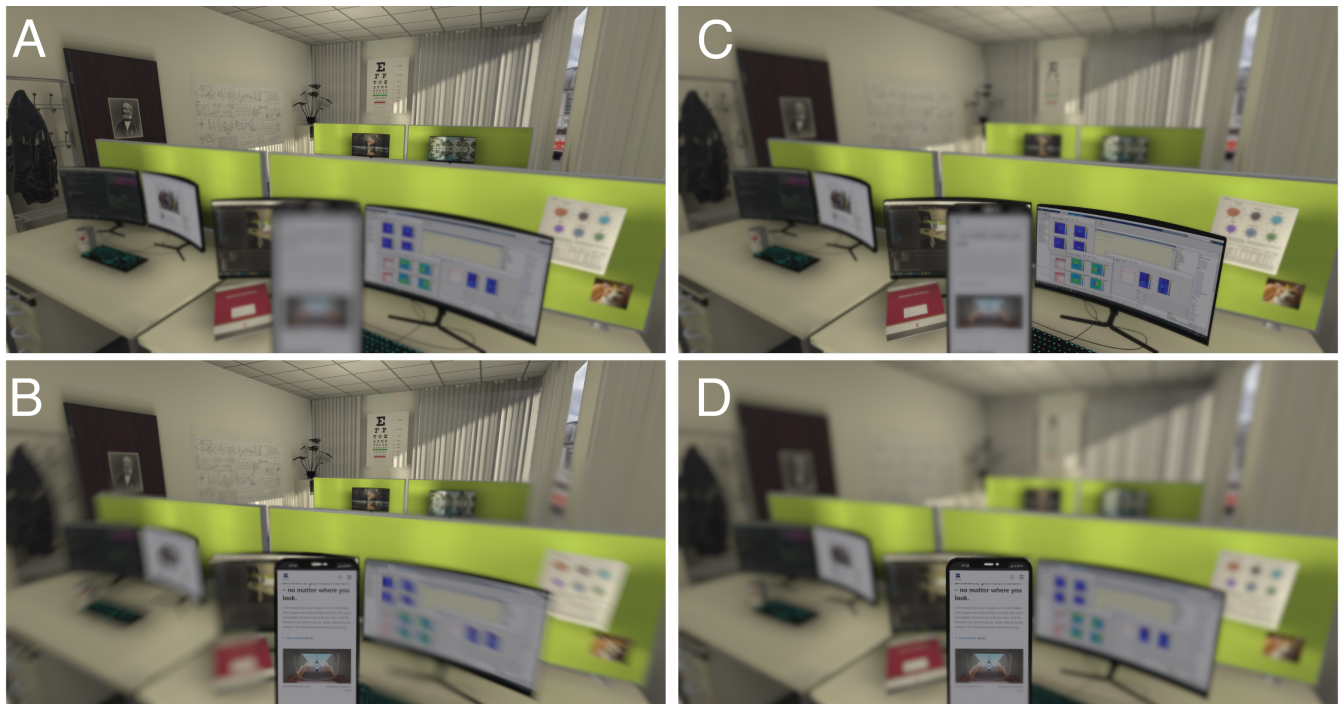


Figure 1: Rendered examples of different simulated ophthalmic lenses. A) Uncorrected presbyopia leads to blurry near vision. The accommodative capabilities of the presbyopic eye are not sufficient to focus on close objects. B) Progressive addition lenses are spectacle lenses for presbyopia, which provide additional optical power in the lower area, to have clear vision for close objects. Peripheral areas suffers from strong astigmatism and will always appear blurry. C) and D) show an autofocusal spectacle lens, which has a tunable optical power to switch focus between different distances. In C) focus is on the computer screen, in D) on the smartphone.

ABSTRACT

Virtual reality is a promising tool to study the influence of ophthalmic lenses on human perception. Independent of the actual refractive errors of the users' eye, vision through different types

of lenses can be simulated while allowing free eye and head movements and easy measurement of behavioural parameters. Especially for presbyopia, existing solutions often lead to visual discomfort. Multifocal spectacle lenses have a spatially varying optical power distribution leading to changing strength and orientation of blurred vision. Understanding the contribution of blur to visual discomfort and studying behavioral changes under the influence of lens blur is a promising application of a spectacle lens simulator. Here we present our framework to simulate spherical and astigmatic blur of corrective spectacle lenses. Size and orientation of a location dependent blur ellipse is calculated live in virtual reality using the depth information of the scene and precalculated power distribution of the lens.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Technical Communications, December 6–9, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9465-9/22/12.

<https://doi.org/10.1145/3550340.3564224>

CCS CONCEPTS

• **Computing methodologies** → **Rendering**; *Virtual reality*.

KEYWORDS

simulation of spectacle lenses, real-time blurring

ACM Reference Format:

Yannick Sauer, Selam Wondimu Habtegiorgis, and Siegfried Wahl. 2022. Realtime blur simulation of varifocal spectacle lenses in virtual reality. In *SIGGRAPH Asia 2022 Technical Communications (SA '22 Technical Communications)*, December 6–9, 2022, Daegu, Republic of Korea. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3550340.3564224>

1 INTRODUCTION

The simulation of optical blur is a long existing topic in computer graphics with many applications from realistic depth of field effects in video games to increasing the sense of reality with HMDs [Barsky and Kosloff 2008; Xiao et al. 2018]. We present a framework which applies post processing blur to simulate vision through spectacle lenses. The major difference to the simulation of depth of field effects is that not only spherical defocus, but also astigmatic aberration, has to be considered, where objects can be blurred stronger along a certain axis. Here we introduce a method to also simulate astigmatic blur with arbitrary orientation. Our method allows for location-dependent aberration strength and orientation, which allows us to simulate realistic blur of uncorrected human vision and different types of ophthalmic lenses. Especially Progressive Addition Lenses (PALs) have strong variation in optical power: the upper area of the lens is designed for far vision and corrects the wearer's far refraction. Towards the lower area of the lens the optical power increases progressively to help focus on near objects. The gradient in optical power will always lead to unavoidable astigmatic aberrations in the periphery, which blurs vision in this area along a location-dependent axis. Another solution for presbyopia treatment are autofocals [Padmanaban et al. 2019], special lenses in early development which can change their optical power to focus dynamically on different distances. Both types of lenses could be improved with further development in their design for reducing discomfort and adapting them to typical human behaviour. To use the spectacle lens simulator as a research environment for such improvements, it is important that the simulation is not based on prerendered images or 3D scenes, but works dynamically using the depth information during runtime to calculate location-dependent blur. Only this way can we allow free movement behaviour in virtual environments that imitate natural depth distribution and measure the influence of the simulated aberrations on behaviour. New optical lens designs can be tested without manufacturing actual lenses, and algorithms for eye tracking-based control of focus distance for autofocals can be developed in a controlled environment.

2 PREVIOUS WORK

Simulations of vision with eye glasses have been implemented in previous work with discussed applications ranging from the optimization of lens design to presentation to spectacle lens customers.

Niesner et al. [Nießner et al. 2012] implemented a real-time simulation of vision through eyeglasses with ray tracing. They used the shape of the front and back surfaces of the spectacle lens

directly as input in their simulation, while in our approach the spherical and cylindrical power is precomputed to easily calculate defocus during runtime.

Barbero and Portilla [Barbero and Portilla 2017] presented a comprehensive spectacle lens simulation including additional chromatic aberration and distortions. Their blur simulation is similar to ours by using a rotated ellipse blur kernel, however, their simulation is strongly simplified by choice of a scene with only one fixed far distance of objects. We instead take the depth distribution into account to calculate blur size for each pixel, and consider possible accommodation of the eye.

3 METHODS

3.1 Spherical and cylindrical defocus

Depth of focus blur is very often rendered using a Gaussian blur kernel, which usually does not produce a realistic effect. The physically correct shape of blur is given by the point spread function (PSF) of an optical system. The PSF in general is anisotropic and varies spatially. For pure spherical defocus, the PSF can be approximated by a disk shaped kernel [Strasburger et al. 2018]. The diameter of this so called circle of confusion is given by

$$b = 0.057 \cdot p_{\text{mm}} \cdot D \quad (1)$$

where p_{mm} is the pupil diameter in *mm* and D the amount of defocus, which can be calculated from optical power P (1 divided by focus distance) and object distance d as

$$D = P - \frac{1}{d} = \frac{P \cdot d - 1}{d} \quad (2)$$

The simulated pupil diameter can be set dynamically in our simulation to simulate blur under different light conditions. Typically the pupil diameter lies between 2 mm to 8 mm [Guillon et al. 2016] with increasing size for lower luminance values. Therefore, blur perceived from defocus is stronger in low light conditions.

Often spectacle lenses correct also astigmatism, where the eye's refractive power is not rotationally symmetric [Read et al. 2007]. A lens with a cylindrical surface is used to correct astigmatism. Only light rays in the meridian along the cylinder curvature will be focused in one point. Rays in the meridian orthogonal to the cylinder axis will not be refracted at all. A more general lens with sphere and cylinder refraction has two defined meridians, along which light will be focused in a single point. To specify such a lens, the spherical power, cylinder power, as well as the axis of the cylinder are needed. The shape of the blur is no longer a circle, rather an ellipse [Blendowske 2015]. The length of the two principle axes is defined by the two power values, sphere and cylinder. The axis of the cylinder determines the rotation of the ellipse. To calculate the exact shape of the blur ellipse, not only one but two defocus values $D_{1/2}$ are calculated using eq. 2 with the spherical power along the cylinder axis and the sum of spherical and cylindrical power along the orthogonal meridian. Given the axis of the cylinder θ , the blur kernel can be described as a rotated ellipse. All kernel sampling points (x, y) have to fulfil the equation

$$\frac{(x \cos \theta + y \sin \theta)^2}{r_1^2} + \frac{(x \sin \theta - y \cos \theta)^2}{r_2^2} \leq 1 \quad (3)$$

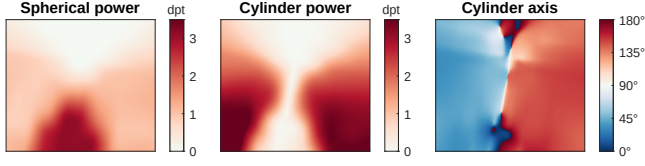


Figure 2: Set of power maps as input of the blur simulation. The spherical power of this example lens is zero in the upper part of the visual field. The optical power increases in the lower part of the lens. The near zone offers 3dpt of additional optical power for focusing close objects. The gradient of optical power always leads to unwanted astigmatism, as visible in the map of cylinder power. The third channel of the input texture contains the information of the location dependent orientation of the cylinder power.

where $r_{1/2}$ are the radii along the principle axes of the blur ellipse given by eq. 1 for both defocus values $D_{1/2}$.

3.2 Location dependent optical power

In general, lenses can have a direction-dependent optical power. PALs, for example, have a progressive increase in optical power. While the upper area corrects the refractive error of the eye for far vision, the optical power increases towards the lower part of the lens to give accommodative support for closer objects. The gradient in optical power will always lead to unavoidable astigmatism in the periphery (blur along a certain direction) [Sheedy et al. 2005]. To simulate this location-dependent optical power, we use maps of the spherical power, cylindrical power and orientation of the cylinder as input. Those power maps are precalculated by the lens manufacturer using ray tracing of the digital surface data of the lens (as described by [Rojo et al. 2014]) and subsequently scaled and cropped by us to fit the dimension of the virtual camera’s image plane. One example set of power maps is shown in fig. 2

3.3 Considering refractive error and accommodation of the eye

Spectacle lenses are designed for the individual refractive error of the wearer and, in case of presbyopes, also for the individual accommodative capabilities of the wearer. Since the correction of the lens is supposed to compensate those errors of the eye, in a simulation of vision through a spectacle lens this correcting power should not lead to blur. We solve this issue by adapting the power map input of the simulation by subtracting the far refraction of specific lens from the power map. The result is a map of spherical and cylindrical error relative to the desired corrective power.

For near vision, the optical power in the near zone of a PAL is only adding additional power D_{lens} to the maximum accommodative power of the eye D_{eye} to help focusing. This power is calculated for a defined near distance d_{near} so that $\frac{1}{d_{\text{near}}} = D_{\text{eye}} + D_{\text{lens}}$. This means that to calculate realistic defocus, the spectacle lens simulation has to also incorporate accommodation. In our accommodation model the eye accommodates to what is necessary for a given object and power of the lens at the objects’ direction up to the maximum accommodative capability D_{eye} . Since a spectacle lens is designed

for a specific value of existing accommodative capabilities, we add this value of accommodative power to the lens power wherever the demand given by the object distance exceeds the power of the lens. This is when the eye would additionally accommodate, but only up to a certain level.

A second possible implementation uses eye tracking to determine the current location of fixation and sets the accommodation for the distance of the object under the measured gaze direction, again taking into account the maximum possible accommodative capabilities. While this eye tracking-based approach leads to more realistic blur in the periphery of the visual field, the result is often not satisfying: because of eye tracking latency and inaccuracy, the fixation distance in complex 3D scenes is not always detected reliably.

3.4 Blur procedure

The procedure of our blur simulation as illustrated in fig. 3 is implemented as a post processing shader in the game engine Unity. In a first shader pass, source render texture and depth map are down-scaled, to perform the actual blurring more efficiently on a lower resolution texture. In the next pass, the two defocus values $D_{1/2}$ are calculated for each fragment using depth texture and power textures and stored independently in the channels of a temporary texture. For the actual blurring we use a predefined disk kernel of fixed size, which samples points around each fragment. Pixels are summed up to the current fragment if eq. 3 is fulfilled for the distance vector between fragment and sample point. If any of the two defocus values is negative, the pixel value is added to a foreground buffer, otherwise to a background buffer. Both buffers are divided by the number of included sample points. Writing in a foreground and background buffer will prevent blur discontinuity. Blurry objects in the foreground will overlap with objects in the background, while blur of background objects will not reach into objects in the foreground. Finally, foreground and background buffers are combined to one blurred texture (with downscaled resolution) before the output texture is interpolated in the last shader pass between source texture and blurred downscaled texture using the blur strength as interpolation coefficient.

4 RESULTS OF DIFFERENT SPECTACLE LENSES

Fig.1 shows how this procedure is applied to the simulation of different spectacle lenses. Setting the input power map to zero simulates only the given accommodative capabilities of the eye influence vision (presbyopic vision). Fig. 1 B shows the simulation of a PAL: through the upper area far objects are sharp, through the lower area close objects are in focus, and the lower left and right periphery present strong astigmatic blur. Another possibility is to change the power map input during runtime, allowing us to simulate focus tunable lenses, which can be used to set the focus dynamically on the current fixation target. Controlling this dynamic change of simulated power by eye tracking could simulate natural depth of field to increase immersion in virtual reality.

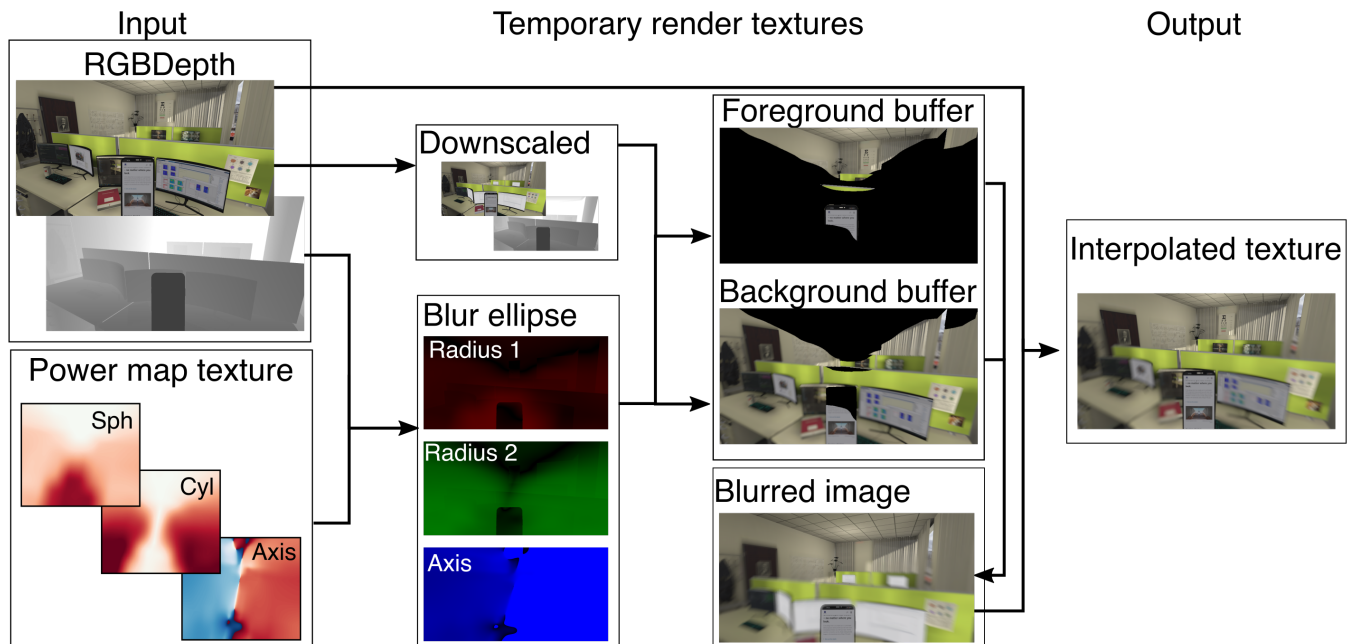


Figure 3: Workflow of the location and depth dependent blur shader: Inputs are render texture, depth buffer and power maps texture. From depth and power maps the parameters of the location dependent blur ellipse are calculated. A downscaled version of the render texture is blurred. Depending on depth, blur samples are added to a foreground or background buffer, which are combined in the end. The final output is interpolated from full resolution render texture and downscaled blurred texture.

5 DISCUSSION

We present a post processing approach for the simulation of the blur of ophthalmic lenses in virtual reality. Spherical and cylindrical blur is applied dependent on location and depth. Our simulation of ophthalmic lenses can be used as a research platform for testing the influence of optical aberrations in a controlled environment. The implementation uses only precalculated power maps as input and requires no additional scene-dependent precalculations.

It is important to note that blur is only one component of the optical aberrations of ophthalmic lenses. Especially PALs show a noticeable amount of optical distortions which also negatively affect visual perception of the wearer [Meister and Fisher 2008]. The blur simulation presented here could be combined with our previously introduced simulation of PAL distortions [Sauer et al. 2022] for a full environment to study discomfort of PAL wearers. Also chromatic aberration can easily be included in the simulation by calculating the blur size independently for the different color channels [Banks et al. 2017].

ACKNOWLEDGMENTS

This work was supported by the German Research Foundation (DFG): SFB 1233, Robust Vision: Inference Principles and Neural Mechanisms, TP TRA, project number: 276693517.

REFERENCES

- Martin S Banks, Steven A Cholewiak, Gordon D Love, Pratul Srinivasan, and Ren Ng. 2017. ChromaBlur: Rendering Chromatic Eye Aberration Improves Accommodation and Realism in HMDs. In *3D Image Acquisition and Display: Technology, Perception and Applications*. Optica Publishing Group, DM2F–1.
- Sergio Barbero and Javier Portilla. 2017. Simulating real-world scenes viewed through ophthalmic lenses. *JOSA A* 34, 8 (2017), 1301–1308.
- Brian A Barsky and Todd J Kosloff. 2008. Algorithms for rendering depth of field effects in computer graphics. In *Proceedings of the 12th WSEAS international conference on Computers*, Vol. 2008. World Scientific and Engineering Academy and Society (WSEAS).
- Ralf Blendowske. 2015. Unaided visual acuity and blur: a simple model. *Optometry and Vision Science* 92, 6 (2015), e121–e125.
- Michel Guillon, Kathryn Dumbleton, Panagiotis Theodoratos, Marine Gobbe, C Benjamin Wooley, and Kurt Moody. 2016. The effects of age, refractive status, and luminance on pupil size. *Optometry and vision science* 93, 9 (2016), 1093.
- Darryl J Meister and Scott W Fisher. 2008. Progress in the spectacle correction of presbyopia. Part 1: Design and development of progressive lenses. *Clinical and experimental optometry* 91, 3 (2008), 240–250.
- Matthias Nießner, Roman Sturm, and Günther Greiner. 2012. Real-time simulation and visualization of human vision through eyeglasses on the GPU. In *Proceedings of the 11th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*. 195–202.
- Nitish Padmanaban, Robert Konrad, and Gordon Wetzstein. 2019. Autofocals: Evaluating gaze-contingent eyeglasses for presbyopes. *Science advances* 5, 6 (2019), eaav6187.
- Scott A Read, Michael J Collins, and Leo G Carney. 2007. A review of astigmatism and its possible genesis. *Clinical and Experimental Optometry* 90, 1 (2007), 5–19.
- Pilar Rojo, Santiago Rojo, Jorge Ramirez, and Inés Madariaga. 2014. Numerical implementation of generalized Coddington equations for ophthalmic lens design. *Journal of Modern Optics* 61, 3 (2014), 204–214.
- Yannick Sauer, Malte Scherff, Markus Lappe, Katharina Rifai, Niklas Stein, and Siegfried Wahl. 2022. Self-motion illusions from distorted optic flow in multifocal glasses. *Isience* 25, 1 (2022), 103567.
- James E Sheedy, Charles Campbell, Ewen King-Smith, and John R Hayes. 2005. Progressive powered lenses: the Minkwitz theorem. *Optometry and Vision science* 82, 10 (2005), 916–922.
- Hans Strasburger, Michael Bach, and Sven P. Heinrich. 2018. Blur Unblurred—A Mini Tutorial. *i-Perception* 9, 2 (mar 2018), 204166951876585. <https://doi.org/10.1177/2041669518765850>
- Lei Xiao, Anton Kaplanyan, Alexander Fix, Matt Chapman, and Douglas Lanman. 2018. Deepfocus: Learned image synthesis for computational display. In *ACM SIGGRAPH 2018 Talks*. 1–2.